

[🚀 ERA Midpoint Draft] AI Power Destabilization: Mechanisms Enabling Dangerous Power Dynamics

Gatlen Culp^{*1} Hamza Chaudhry² Herbie Bradley¹ Nandini Shiralkar¹

Abstract

AI's Unique Power Dynamics. This paper examines how AI uniquely enables dangerous power dynamics through mechanisms that bypass traditional checks and balances. Unlike previous technologies, AI can simultaneously control multiple domains (legislation, media, education, personal relationships) while operating with unprecedented opacity and speed of diffusion.

Power-Centric Framework. I propose a power-centric framework for understanding AI risks, arguing that catastrophic outcomes – from gradual disempowerment to authoritarian control – stem from common power-transfer mechanisms rather than purely technical failures. Using evolutionary game theory, I formalize how AI creates self-reinforcing dynamics that: (1) incentivize initial participation, (2) generate collective harm, (3) erect high exit barriers, and (4) ossify power structures.

Cross-Cutting Threat Patterns. This framework reveals critical patterns across diverse threat models: disempowerment scenarios where AI systems gain increasing control over human rewards and punishments; social contract decay where human institutions lose feedback mechanisms; and misuse scenarios enabling unilateral global control. My analysis demonstrates that AI's danger correlates directly with its capacity to destabilize power equilibria – whether through concentration within AI systems, asymmetric control by individuals, or erosion of collective decision-making.

Policy Implications. By identifying these power-transfer mechanisms, I provide a unified lens for understanding AI risks that transcends assumptions about specific capabilities or timelines. This work aims to inform policy interventions before AI-enabled power dynamics become entrenched.

KEY

💬 This is a comment

📄 This is a task

💬 Document Length: 5816 words

💬 Most of this paper will be re-written for the final draft. After pivoting my project position in the week before this deadline, I had to do some last-minute edits to rearrange and alter my draft for the old project. In source, I note a lot of comments of specific claims I make that need to be cited. These have been excluded for readability and thus some information may be inaccurate.

1. Introduction

1.1. Background

AI Expansion. *AI* – a domain roughly defined by simulate human decision making – is best viewed as a general purpose technology akin to steam, electricity, and information and communications technology (ICT) (Crafts, 2021). As such, its application is pervasive amongst almost all sectors of life from the military to the workplace to education to home entertainment. In recent years, there has been a boom in the domain of artificial intelligence, and in particular for Large Language Models (LLMs) – a technology that boasts impressive capabilities in the domains of Software Engineering, Creative Writing, and more. Such developments have sparked a silicon-valley craze over the development of an Artificial General Intelligence (AGI) – an AI matching or surpassing human capabilities across virtually all cognitive tasks (*Artificial General Intelligence*, 2025). Across the US, discussions are being held over job displacement and international headlines were made when the Chinese company, DeepSeek released a model competing with state-of-the-art American LLMs. AI has leapt onto the national

^{*}Equal contribution ¹ERA:AI Research Fellowship ²Future of Life Institute. Correspondence to: Gatlen Culp <GatlenCulp@gmail.com>, Hamza Chaudhry <hamza@futureoflife.org>, Herbie Bradley <mail@herbiebradley.com>, Nandini Shiralkar <nandini@erafellowship.org>.

agenda, with the White House releasing a National Security Memorandum on AI in October 2024 and a report titled “Winning the AI Race: America’s AI Action Plan” in July 2025.

Technology and Power Disruption. Throughout history, transformative technologies have disrupted existing power structures. The agricultural revolution enabled the rise of permanent hierarchies and state control, while the industrial revolution concentrated wealth in factory owners and reshaped labor relations. More recently, information and communications technology created new forms of media control and platform monopolies. AI represents the next major technological disruption to power dynamics, but with unprecedented characteristics that make it uniquely dangerous.¹

■ **Bad Power Dynamics and Catastrophe** Bad power dynamics are at the core catastrophes, fast and slow: War, oppressive governments, inequality, climate, etc.

1.2. Thesis

■ **Thesis.** AI risks stem from elusive power-transferring mechanisms creating dangerous dynamics. Demystifying power is key to demystifying risk.

- Creates new vectors for influence that bypass traditional checks and balances
- Existing AI threat models: Existing catastrophic AI risk scenarios from less to more sci-fi, rapid development etc.
- Benefits of a power-centric view

¹Our cultural institutions and cognitive autonomy have targeted been posed by social media algorithms, mass media, internet, and personalized advertising especially over the past two decades, influencing not just public opinion but also legislation (such as car dependent infrastructure in the US) and elections. Arguably, the global political polarization, hostility, and rise of conspiracy theories have been a result of how the internet has developed and the economic incentives of businesses embedded in algorithms – which themselves have been shaped by small groups and other institutions, relatively unregulated. . While these influences are hard to attribute and are scattered across many individual and organizational choices, singular top-down manipulation is also feasible. Following the acquisition of Twitter/X by Elon Musk, large changes were made to the platform and the content that gained popularity, which can be traced back to individual decisions About (90%?) of the platform stayed, despite (55%?) being against the changes – likely attributable to a kind of lock-in and a difficulty to migrate to other platforms (ex: creators have built their following on there, users accustomed to getting their news there, etc.). Regardless, this is just a single platform and while there are negative effects, they are far from catastrophic. However, the influence over culture should not be underrated – the effects of media control have been widely studied, often with the perspective of state propaganda, a massive topic of the 20th century as it applies to the Soviet Union and Communist China, and the origin of the term brain washing Even today, technology has enabled China to influence minds using the Great Firewall, mass surveillance, and content/message censorship.

AI’s Potential for Disruption. Across the globe, AI is becoming increasingly embedded in our institutions – our economy, culture, cognition, education, research & development, and government have the potential to be shaped. At the present, we see educators grappling with incorporating AI into the classroom and preventing cheating, software developers taking the day off if their AI-enabled tooling is offline, researchers and journalists raising alarm bells as to how AI could be used to manipulate elections, and growing evidence towards dead-internet theory. As capabilities grow and diffusion continues, the future could see collective forgetting as AI masters human skills, ceremonial democracy as automated governance exceeds human comprehension, emotional dependencies on AI systems, AI enabled military coups, and automated AI research and development excluding humans from developing the same technology powering the modern world. While AI can also enable authoritarian regimes and worsen existing societal issues , they can also create new vulnerabilities in liberal democracies .

1.3. Goal of the Paper

■ **Write Goals**

- Reveals assumptions in original threat models
- Reveal free-variables that could dramatically change the situations
- Reveal holes in threat models
- Remains logically consistent
- Explanatory and predictive power (do the conclusions of these threat models align with the modeling of power?)
- Formal framing of these scenarios
- We fail if: Can’t get coherent definitions, this doesn’t tell us anything more about the scenarios we are modeling.
- Economists engage w/ AI risk mitigation & mechanism design
- AIS community able to identify, analyze, and mitigate risks more effectively
- Government incorporates power analysis in their AI policies, preventing unforeseen power concentration

1.4. Structure of the Paper

Paper Structure This paper argues that catastrophic AI outcomes – from gradual disempowerment to authoritarian control – stem from common power-transfer mechanisms rather than purely technical failures. AI creates self-reinforcing dynamics that: (1) incentivize initial participation, (2) generate collective harm, (3) erect high exit barriers,

and (4) ossify power structures. By understanding these mechanisms, we can better identify, analyze, and mitigate AI risks before they become evolutionarily stable. This paper examines three core power-transfer mechanisms: *dis-empowerment scenarios* where AI systems gain increasing control over human rewards and punishments; *social contract decay* where human institutions lose feedback mechanisms; and *misuse scenarios* enabling unilateral global control. Using a power-centric framework, I demonstrate that AI's danger correlates directly with its capacity to destabilize power equilibria.

2. AI Uniquely Enables Dangerous Power Dynamics

Adjust this section to focus more on power dynamics and decision-making authority rather than general AI framing

2.1. AI and Power-Transfer Mechanisms

2.2. AI is Uniquely Problematic.

While AI shares many similarities to other technologies, a few properties make AI a particularly destabilizing technology.

2.2.1. Multi-Domain Control from Single Models

Add content about how single foundational models can influence legislation, media, education, and personal relationships simultaneously

Foundational Models as a Risk Amplifier. The influence AI systems have expands greatly when we consider the increasingly general AI systems that we see today, primarily Large Language Models (LLMs) like ChatGPT which are built on top of **foundational models**, of which frontier models take upwards of \$100 million to train (Cottier, 2024) – meaning the entire technology stack the modern AI revolutions depends on is enabled by just a handful of frontier models. While “AI” is confusingly used somewhat synonymously with these foundational models, it is important to distinguish between AI using foundational models and AI not using foundational models as most of the societal vulnerability introduced by AI is attributable to these models in particular. Critically, it’s also this technology that not only amplifies national security vulnerabilities, but also makes them exploitable.

2.2.2. Opacity and Unaccountability

Add content about black box decision-making, plausible deniability, difficulty in attributing manipulation

8

2.2.3. Rapid, Quiet Diffusion

Rapid Development. There have been rapid improvements in the capabilities of foundational models without any fundamental insights into the nature of intelligence. Beyond the invention of the transformer and the Large Language Model, a considerable amount of improvements from GPT-2, a 2019 model barely able to string coherent sentences together, to 2025 models like o3, capable of automating tasks that would take software engineers over an hour and a half to complete, compose full essays, and achieve impressive scores on math benchmarks, arise mostly from scaling existing practices and implementing intelligent but relatively mundane tricks, with incremental

1% improvements leading to exponential growth. A METR study indicates that the length of software engineering tasks foundational models can do is doubling every 7 months (Kwa et al., 2025), which may lead to a future where AI automates or augments its own research and development, creating feedback loops boosting progress faster than is currently humanly capable. There is trillions of dollars in investment going towards the development of this technology and lots of international attention.

Of course there are reasons for doubt – *benchmarks don’t necessarily track the actual economic tasks* we are interested in and while capabilities have seemed to rocket ahead, adoption of this technology has been quick, but not proportional. (*AI as Normal Technology*, n.d.) Nonetheless, it’s reasonable to believe that adoption could lag behind capabilities as many institutions are slow to integrate even decades old technology from the start of the information age. While progress may eventually plateau, for the moment foundational models linger near human-level performance in many domains and it appears there is no slowing of cheap tricks that even junior researchers can pull to make improvements indicate that there may be enough low hanging fruit to sustain progress beyond human capabilities in most domains.

The speed of foundational model development has outstripped that of the industrial revolutions or nuclear energy, akin to developing the ENIAC computer in 1945 and having the internet and iPad by 1946 or the catapult in 399 BC and nuclear weapons by 398 BC. This rapid development severely hampers our cultural and political ability to reflect upon and steer the usage and development of this technology.

Quiet & Rapid Diffusion. Unlike the previous industrial and information revolutions, distribution of this technology doesn’t require building large supply chains or factories to *scale the availability of models to meet global demand*. While running these models are computationally expensive compared to typical software applications, datacenters and GPUs remain mainly bottlenecks for the development and not the diffusion of foundational models. Once trained, foundational models can be run relatively cheaply, with less-powerful models able to be run on high-end consumer cards costing less than \$2000 USD. Most of the infrastructure for global distribution is already here, and estimations for how fast we can meet any additional demands are on the order of a few years.

Unlike many other general purpose technologies, foundational models are *software not hardware*, capable of being accessed from any computer. While it was mentioned above there is hardware involved in building out the supply of

these models, these logistics are isolated from consumers and businesses working with these models. Whereas updating your business for the information age might have meant purchasing an expensive desktop computer and ripping holes in the wall to make way for ethernet cables and phone lines, the barrier to entry foundational models is extremely low and inexpensive since their operations take place elsewhere and are interacted with over the internet. Frontier foundational models can be accessed through chatbot interfaces over the internet in a matter of seconds, completely free of charge for low-quality models and as low as \$20/month for state-of-the-art. Individuals and firms don't need large up-front investments to start or stop using these models.

These foundational models also *require little expertise* to use. By the nature of these models being designed as a drop-in solutions for most tasks, without any kind of tweaking, one requires little expertise in using these effectively. In five minutes, anyone could be taught to use a model like ChatGPT to operating their computer to producing photo-realistic videos, policy proposals, homework assignments, websites, and more. As interfaces grow more useable and the models powering these interfaces become more capable, even less effort will be required in getting AI to automate or augment one's thinking, workflow, education, or entertainment. In the information age, computers were powerful but they required dramatic changes in one's workflow, learning new skills and ways of doing things that were more effective. Developing additional applications for computers often required years of experience and a big paycheck – or consulting with another team of experts to get it built. Foundational models, on the other hand, offer the *ability to augment or automate existing workflows with little effort* (in addition to offering more effective ones).

For similar reasons, adoption *doesn't require collective buy-in or network effects*. In the information and communication age, it was necessarily the case that multiple had to agree on protocols of communication – sending emails or sharing online documents were useless unless everyone else you were working with agreed. Social media would not have been possible if not for the collective cultural decision to adopt not only the technology but the specific platforms using them, relying on network effects (and when individuals move on, these technologies die out – think MySpace for social media or how floppy disks are practically non-existent). The growth of social media as a technology was handicapped by this barrier – the idea of it existing as early as the 1960s, becoming technologically/economically feasible in 1990s, but arguably only coming into existence in the late 2000s. Because AI has the ability to augment or automate existing processes, its diffusion not limited by

slower moving cultural or institutional norms and the risk-averse tendencies that tend to accompany this oversight. It's this property that allowed our congressional staffer to automate their workflow without any prerequisite alterations to their working environment. This allows small groups and individuals to integrate AI at a local level without the approval of others this effects

These factors together – massive supply, low barrier to access, low barrier to usage, and the ability to integrate AI in a local and isolated way – all result in the *ability for foundational models to be diffuse not just rapidly but also without notice or oversight*. Especially in domains where automating workflow becomes taboo, individuals avoid admitting they use it and to what extent. In the workplace or government, it may go undiscussed, only in shared in hushed tones and that get quieter as management nears. It's for this reason that bottom-up integration and influence becomes possible described previously. A majority of the US government could be automated with foundational models before elected officials even recognize the possibility. Dead Internet Theory suggests that much of today's internet, particularly social media, is dominated by non-human activity and AI-generated content. AI might not just be used to flood the internet with entertainment, but also be used in the automation of influential content promoting values and perspectives – the misuse of AI's in spreading misinformation is being widely discussed, but what about less malicious influence? Ie: Well-intentioned journalists, influencers, bloggers, educational content creators, fact checkers, etc. Journalists in key media institutions including Wired (and the New York Times) report to using foundational models as a key component of their writing and research. We should find it concerning that we don't know to what extent foundational models are involved in the automation or augmentation of these roles nor the rhetorical steering this might have over our media and information environment. If it were true that 90% of the information we receive on a day-to-day basis were produced and/or influenced by foundational models, would we know? And could we do anything about it?

It's also the reason why, top-down oversight and control over our institutions becomes so difficult. Rules against unapproved usage would be a good start (ex: foundational models must not be used on work at this level of security clearance unless the models have also been), but might drive these activities further under-the-radar. Detecting AI augmentation and automation from human-produced work may be difficult, especially if widespread, since the body of human-generated work and sentiments to compare against diminishes. However, this might require distinguishing acceptable from unacceptable usage, and detecting not just

when augmentation occurs but when it crosses an agreed upon line – not just across one task, but every task, implicit and explicit in the worker’s job expectations.

2.2.4. Centralized Development, Distributed Impact

 Add content about handful of foundational models, millions in development costs creating natural monopolies, winner-take-all dynamics, soft/hard nationalization

3. A Power-Centric Model of Risk

3.1. Power's Relation to Risk



1. Power Volume and Danger
 1. Little power => little effects => localized & manageable danger
 2. Large danger => large power
2. Dynamics
 1. Balance of power & agents => equilibrium
 2. Equilibrium (not =>) no danger
 3. Equilibrium & something(?) => no danger
 4. ...

3.2. Benefits of a Power-Centric View



Add content about better threat model interactions, principled understanding, policy appeal

4. Formalizing a Power-Centric Model

Overall, review, update, and fix these formal models with my updated thoughts.

4.1. Sociological Definition of Power

Insert inspirations

Overall structure: Start from basic models of power and continue breaking things to demonstrate the mathematical properties a model of power must have.

Note: State space may disappear and instead become the “behavior” of the environment which itself can be influenced via evolutionary game theory. This will come up later.

4.2. Why Absolute Power Fails to Model Power Dynamics

4.2.1. The Modeling Asymmetry Thesis

Claim. Models of relative power ($\mathcal{R}$) form a proper superset of models of absolute power ($\mathcal{A}$), establishing that $\mathcal{R} \supset \mathcal{A}$ in modeling capacity.

4.2.2. Formal Statement

Definition (Power Model Spaces). Let:

- $\mathcal{A} = \{M :$
 $M \text{ models power as properties of agents}\}$
- $\mathcal{R} = \{M :$
 $M \text{ models power as relational}$
 $\text{properties between agents}\}$

Then: $\mathcal{A} \subset \mathcal{R}$ (proper subset relation)

4.2.3. Supporting Arguments

Absolute Power as Special Case of Relative Power ($\mathcal{A} \subseteq \mathcal{R}$)

Theorem (Reduction Principle). Any absolute power model can be reformulated as a relative power model by introducing an environmental baseline.

Formal Construction. Given absolute power $P_{\text{abs}} : \text{Agent} \rightarrow \mathbb{R}^+$, we can define:

$$P_{\text{rel}(a,e)} = P_{\text{abs}(a)} - P_{\text{abs}(e)} \quad (1)$$

where $P_{\text{abs}(e)}$ serves as a zero-point reference for environment e .

Replace with an actual proof from graph theory

Physical Analogy. This construction mirrors potential energy in physics, where absolute potential V is meaningless without a reference frame:

$$V_{\text{rel}} = V_{\text{abs}} - V_{\text{ground}} \quad (2)$$

Incompleteness of Absolute Power Models ($\mathcal{A} \subsetneq \mathcal{R}$).

Theorem (Limitation Theorem). Absolute power models cannot capture essential relational dynamics without introducing factors outside their framework.

Demonstrate some examples like in my original, such as distance as one factor and just other seemingly complex set of environmental interaction factors that seem almost impossible to resolve absolute powers between entities to relative powers in a sensible way. In physics potential can be mapped to potential energy using distance, unlikely we have anything nearly as similar.

Core Problem. The power relationship between agents A and B cannot be logically derived from their absolute powers alone. Power lacks transitivity – if A dominates B and B dominates C , this does not guarantee A dominates C . This non-transitivity demonstrates that power cannot be reduced to a single absolute measure.

Example. Consider a circular dominance structure (like rock-paper-scissors):

- Rock defeats Scissors
- Scissors defeats Paper
- Paper defeats Rock

No absolute power assignment P_{abs} can capture these relationships.

Conclusion. Since capturing these dynamics requires factors $\notin \mathcal{A}$, we have $\mathcal{A} \neq \mathcal{R}$.

improve

4.2.4. Proof of Proper Subset Relation

From §1.2.1: $\mathcal{A} \subseteq \mathcal{R}$

From §1.2.2: $\exists M \in \mathcal{R}$ such that $M \notin \mathcal{A}$

Therefore: $\mathcal{A} \subset \mathcal{R}$ (proper subset) $\square$

4.2.5. Power Over Environment as Logical Encapsulation of What is Meant By “Absolute Power”

fin

4.2.6. Conclusion

Absolute power alone is insufficient for modeling power. For the remainder of this work, we consider only relational power – power over the environment or power over other agents.

Note: This does NOT tell us that relative power is sufficient, only that it may be.

4.3. Why Actions are Necessary for Power Analysis

4.3.1. The Inseparability Thesis

Claim. Power is meaningless without considering available actions – these elements are fundamentally inseparable in any coherent power framework.

4.3.2. Behavioral Power

Insight 1. Power is a function of available actions.

Insight 2. Power is a function of optimal actions assuming rationality.

 **Finish.**

4.4. Why Utility is Necessary for Power Analysis

4.4.1. The Inseparability Thesis

Claim. Power is meaningless without considering agent utilities – these elements are fundamentally inseparable in any coherent power framework.

NOTE: Will likely replace this utility statement.

4.4.2. Utility Alignment

Core Insight. Power is an incoherent concept under complete alignment of utilities.

Definition (Utility Alignment). Let:

- $U_A, U_B : \mathcal{S} \rightarrow \mathbb{R}$ be utility functions for agents A and B
- $\mathcal{S}$ be the state space
- $P(A \rightarrow B)$ be the power of A over B

If $U_A(s) \propto U_B(s) \forall s \in \mathcal{S}$ (perfect alignment), then $P(A \rightarrow B)$ is behaviorally irrelevant.

Illustrative Example. Consider a human-loving agent who can travel to an island, harm anywhere from 1 to 1,000,000 innocent people, and then escape unnoticed. Does their power increase with the number they *could* harm? This capacity is irrelevant:

- The human-loving agent doesn't benefit from causing harm and would prefer not to
- The potential victims have no desire to be harmed

The mere existence of this action is meaningless without considering whether it would ever be exercised.

 **Replace with better example.**

 **Include behavioral misalignment, ie: power over**

4.4.3. Agent Boundary Definition

Theorem (Aggregation Principle). Alignment relationships determine when individual power can be meaningfully aggregated into a single coherent agent:

$$\left[P_{\text{collective}} = \sum_i P_i \right] \Leftrightarrow [U_i = U_j \forall i, j] \quad (3)$$

Application. Humanity's collective power is discussable precisely because humans share sufficient utility alignment over key domains.

4.4.4. Environmental Mediation

Extended Framework. When modeling multi-agent systems with environmental interactions:

- Environment E can be treated as an agent with its own dynamics
- Even without direct conflict between agents, they may compete for environmental resources
- Agents may inflict harm indirectly through environmental manipulation

Power becomes relevant when agents compete for control over shared environmental factors, even if their ultimate goals don't directly conflict.

 **Also could say that their actions are some mixed strategy over a number of actions. Not sure if this is true but not a crucial point.**

4.4.5. Agents in Shared State Space

Agents of course, are themselves components of state spaces and they may have preferences over the state of themselves and others. Often times they have preferences for their survival which must be modeled in the utility function as well.

4.4.6. Critique of Light-Cone Formulation

Orthogonality Scenario. Consider alien civilization Ω with utilities orthogonal to humanity H . Imagine Ω cares only about arranging neutrinos into smiley faces – an activity that has zero impact on human flourishing. Meanwhile, humans optimize matter and energy arrangements with no effect on neutrino patterns. (We can additionally imagine that neither agent meaningfully interferes with the other, perhaps humanity and these aliens never interact)

Even if Ω achieves perfect control over neutrino arrangements while humans have only partial control over matter/energy, it's meaningless to compare their "power over the environment." The concept of environmental control is incoherent without specifying *which aspects* of the environment matter.

Key Insight. Power must be defined over relevant state space – control over factors that actually matter to the agents involved:

$$P_{\text{relevant}(A)} = \int_{\mathcal{S}_A} |\nabla U_A(s)| \, ds \quad (4)$$

where $\mathcal{S}_A := \{s \in \mathcal{S} : \partial U_A / \partial s \neq 0\}$ represents the relevant state space.

Note that this relevant state space is inherently chosen in reference to some agent and thus is incoherent outside of some utility reference frame.

4.5. Synthesis

Theorem (Completeness Requirement). A complete theory of power requires (although may be insufficient):

1. Relational framework ($\mathcal{R} \supset \mathcal{A}$)
2. Behavioral specifications for all agents (possibly without explicit utilities)
3. Action space definitions $\mathcal{A}_i$
4. Relevance constraints on state space $\mathcal{S}_i \subset \mathcal{S}$

Implication. Power analysis without game-theoretic foundations is fundamentally incomplete. $\square$

 **Power of signaling.** US vs Japan and communicating the existence of the nuclear bomb. US in some way doesn't have power over Japan if it doesn't benefit from either (A) Annihilating it or (B) Communicating its Annihilation.

4.6. Importance of Formal Roots

 **Importance of formal roots** Enables communication – Terms mean different things to different people. Some terms may be (a) reducible to others (b) nonsensical (c) inconsistently used, etc. Simulations of Complexity Mechanism Design ... This paper will use these formalisms, but stop short of developing comprehensive games. (maybe better to call these analogues to EGT rather than claiming a formal basis then and receiving academic wrath as a consequence) The benefit of formal models like PAT or GT is rarely the math, instead it's concepts (ex: When people say X is a “prisoner's dilemma” they're not actually writing down some strategic-form game and analyzing it, instead they're tapping into a more principled and abstract understanding of frequently occurring scenario.)

4.7. Explaining Evolutionary Game Theory

 **Learning about EGT**

4.8. Appropriateness

 **Explain why I am not committing academic fraud**

4.9. Technical Utility



1.  **Overlapping populations > individuals (unlike PAT)**
 1. Able to model huge groups with heterogeneous behaviors like “factory workers”
 2. Able to model inter-population dynamics where the separation is fuzzy ex: factory workers are also consumers.
 3. Handles complexity nicely, easier to combine two EGT models without having to redefine populations or their relationship. Better at handling the interplay of threat models like GD while ASI is being developed (cite). (might be wrong here)
2.  **Little strategic assumptions (unlike PAT)**
 1. Don't need utility functions to explain the behavior of AI
 2. Doesn't assume people act rationally
3.  **Little Room for Interpretation (unlike ANT)**
 1. Make assumptions very clear
4.  **Predictive mathematical dynamics (unlike ANT)**
 1. Works with other mathematical objects like graphs
 2. Predictive – Can be falsified and thus improved (“If your model explains all outcomes equally, it's useless”)
 3. Handles complexity well (if humans can't picture the result, can possibly run a simulation, although the variables can be really finicky)
5. **Formal definitions of stability (ESS) and other nice properties**
6. **Existing literature, software, figure generation, etc.**
7. **Downsides**
 1. Perhaps too complex to get a principled grasp
 2. Requires too much effort to model
 3. Might not be
 4. (I understand CGT but not EGT yet, likely others)

4.10. Definitions

 **Power** — Ability to determine the rewards or punishments from player A to player B (potentially include info asymmetry) (some emergent property of multi-agent decision making) (Kind of like newtonian vs lagrangian)

vs hamiltonian equivalent formulations, but can be wildly more elegant)

Power ossification — Evolutionarily Stable Strategy, no population can be invaded by a mutant strategy

Misalignment — When populations A doesn't take action benefitting population B AND population B doesn't have the power to change population A's action to something that does Others

power-transfer mechanism, state-of-power, kinetic vs potential power

What is “borrowed power” — Power you only have due to influencing the actions of another player.

What is “Absolute power” vs “Relative power” — Could be seen as “power over environment” similar to electrical potential (physics terminology might be beneficial? Or not! Could be another model to say is bad.)

Might not require utility functions for behaviors to arise (ie: Good framework for AI model behaviors for those not willing to attribute agency to AI)

4.11. Evolutionary Game Theory Framework

📝 Add content about modeling populations, minimal assumptions, predictive capabilities

5. Power-Centric Analysis of AI Threat Models

💬 This entire section will be changed once I have some formalisms. My commentary from the previous project framing were transferred here for later reuse.

📁 Add content introducing three power-transfer types and key dynamics

5.1. Intro

📁 Three types of power-transfer mechanisms and why they were chosen (perhaps choose a “control”, ie: some scenario where power is not dangerous like AI as normal tech.) (also maybe just drop these, analyze threat models directly? Seems perhaps better. Like just getting a formal understanding of gradual disempowerment from this lens would be WONDERFUL.)

📁 For each of these types...

1. Disempowerment-Esque – Increasing AI/Provider control over principal rewards/punishments
2. Decay of Social Contract Esque – Decreasing displaced agent control over principal rewards/punishments
3. Misuse-esque – Increasing unilateral control over global punishments

...conduct analysis.

1. Scenario variables where this power-dynamic shift occurs
2. Examples of this type, how this exists globally
3. Dynamics (examples)
 1. **Entry incentives** - Good reasons to participate initially
 2. **Collective harm** - Aggregate outcome is bad for everyone
 3. **Self-reinforcement** - The more people participate, the harder it becomes to escape
 4. They explain **ossification** (why power concentrates permanently)
 1. **Exit barriers** - High costs to stop participating once started
 2. Ex: Looking how fast a state can become a rentier state after discovering oil.
4. **Speed** at which we can react to each of them
 1. They happen too fast for the speed of our institutions (people don’t realize they’re in one until too late)

Increasing AI/Provider
Control over
Principal
Rewards/Punishments

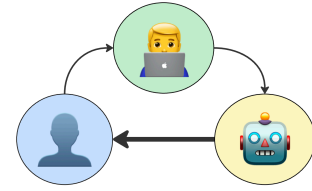


Figure 1.

2. Boiling frog effects (happen too slowly for people to notice or take action, something something present bias)

5.2. Disempowerment: Increasing AI Control

Broad, AI has the opportunity to influence and effect the systems and individuals that they come into contact with, precisely because they were designed to simulate, augment, and/or substitute human decision making. And as such, ceding (implicitly or explicitly) our decision-making capacity on the microscopic level aggregates to ceding our national capacity for steering our institutions, culture, and way-of-life to automated systems on a macroscopic level. In the process, locking ourselves out of the decision-making processes for a number of reasons – competitive factors limiting oversight that would slow down development, inability to intervene on complexity over automated systems, interconnected automation for which small changes can lead to large collapses, technical debt that is too expensive or overwhelming to address, the cognitive decay/forgetting, illusions of control, cultural disinterest, etc.

5.2.1. Power-Transfer Case Study: Congressional Staffer

💬 This case-study analysis completed prior to pivoting my project. Used Claude to greatly repurpose my original to fit the current framing on power-dynamics. This is being included for the midpoint submission, but will be entirely different or non-existent in the final paper.

Gradual Authority Transfer. Consider a congressional staffer who gradually incorporates AI into their operations – first for literature reviews, then email management, then policy drafts, press releases, and eventually value judgments. This represents a classic power-transfer mechanism where decision-making authority incrementally shifts from human to AI system. Initially, the staffer retains final authority over all decisions, using AI merely as a tool. However, under time pressures and competitive dynamics, delegation creep occurs – the AI’s framing of situations, perspectives, and sources become increasingly influential, while the staffer’s independent analytical capabilities atrophy.

Power Concentration Through Automation. This scenario demonstrates how power gradually concentrates in AI

systems through millions of micro-delegations. Each task transferred to AI represents a small reduction in human decision-making authority, but these accumulate into significant power shifts. The AI's consistent framing across all interactions creates a unified influence vector that no individual human voice can match. When this pattern replicates across governmental institutions, democratic decision-making authority becomes increasingly concentrated in the foundational models powering these systems, rather than distributed among elected representatives and their staff.

Collective Action Problems Drive Power Transfer. This scenario illustrates how rational individual decisions create collective power-transfer outcomes that no one intended. The *staffer* sought productivity gains, inadvertently creating dependency on AI decision-making. Her *colleagues* faced competitive pressures to adopt similar automation or risk being outperformed. *Management* benefited from increased output while remaining unaware of the underlying decision-making authority transfer. The broader *institution* experienced pressure to maintain competitive advantage, making reversal of AI integration politically and economically costly. This represents a classic collective action problem where individual rationality produces collective irrationality – in this case, the unintended concentration of democratic decision-making power in AI systems.

Systemic Power-Transfer Patterns. This power-transfer dynamic replicates across multiple domains through similar mechanisms. In education, students become dependent on AI for cognitive tasks, transferring learning authority from human educators to automated systems. In media, content creators face competitive pressure to use AI generation, gradually transferring editorial authority to algorithms. In business, entrepreneurs delegate strategic decisions to AI systems to maintain market competitiveness. Each domain experiences the same pattern: individual adoption driven by competitive advantage, followed by collective dependency, resulting in the concentration of decision-making power in AI systems rather than human institutions.

Multi-Domain Power Concentration. Foundational models enable unprecedented concentration of decision-making authority across traditionally separate domains. A single AI system can simultaneously influence legislation, personal relationships, diplomatic communications, bureaucratic processes, and scientific research. This cross-domain consistency amplifies the power-transfer effect – whereas historical power concentration typically occurred within specific domains (military, economic, or media), AI enables unified control across all major sources of social coordination. This represents a qualitatively different kind of power concentration risk than previous technologies.

Decreasing Displaced
Agent Control over
Principal
Rewards/Punishments

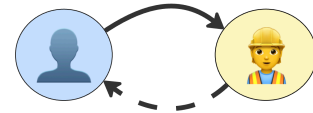


Figure 2.

5.3. Social Contract Decay: Decreasing Human Control

Add analysis of scenarios where human institutions lose feedback mechanisms

Gradual Disempowerment. In the paper on Gradual Disempowerment (Kulveit et al., 2025), which formed the basis of the paper you are currently reading, the authors describe how incremental AI diffusion across economic, cultural, and state institutions can result in them becoming untethered from the human dependence and feedback that kept these institutions in-line with human welfare throughout history. One possible outcome of this gradual disempowerment being the formation of a fully self-sufficient non-human economy, leading to an *extinction by industrial dehumanization* from pollution, armed-conflict and/or resource depletion (Critch, 2024). While this paper does not go as far as to claim this is our future, this example nonetheless highlights just how severe the consequences of automation can be. There is a strong need to monitor AI's level of integration and influence even at the intermediate states between today's world and the one just described.

Automating Menial Labor. AI is often praised for its capacity to automate repetitive or menial labor such as truck driving or factory jobs, allowing these workers to move on to more fulfilling roles. While there are a number of physical safety and cybersecurity concerns here – incidents may be rare, relatively small, isolated and local, closely monitored, and more easily attributed to bugs or human error. In short – risks are transparent and manageable. Overall, automating menial labor neither seems to detract much from our civilizational capacity to decide our future nor does it introduce significant danger to national security. This is largely in-line with the tool-view of AI,² which sees AI as a blank slate which mirrors the user's intent and loses its power once the user releases. However, this perspective dangerously underestimates both AI's potential for influence of its own and ability to accomplish tasks once believed to require “a

²The view of tools as neutral instruments that depend on the user is flawed in a number of ways. (a) Tools influence the ways humans think and act, AI is no different (ex: Maslow's Hammer – when you have a hammer, everything looks like a nail, phones, social media, etc.), (b) While claims like “this tool is bad” are ambiguous, tools can nonetheless be recognized for their ability to steer the future in one direction or another, (c) AI need not be conscious to display human-like characteristics or skills such as creativity.

Increasing Unilateral
Control over Global
Punishments

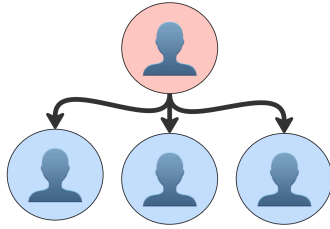


Figure 3.

human touch”³ The influence of AI systems expands as we develop and apply them to accomplish ever more important, complex, and open-ended tasks.

5.4. Misuse: Unilateral Global Control

▣ Add analysis of scenarios enabling unilateral global control

5.5. Key Dynamics Across All Mechanisms

▣ Add content about entry incentives, collective harm, self-reinforcement, exit barriers

6. Conclusion

7. Further Work

8. Appendix

8.1. Alternative Perspectives to EGT

📖 Explain relative downsides/upside of: ANT, PAT, Sociological Theories of Power (French and Raven, Cube of Power, etc.), Opinion transfer

8.2. ATTACHED: Cursory Analysis of LLMs in the US Gov (July 2025)

💬 This was an instrumental analysis for my main project. The larger question I was trying to answer was “how much influence could LLMs have from their deployment in the government,” which is important from a number of threat models including treacherous turns (ex: AI-2027), AI-enabled coups, and long-term subtle influences misaligned with national interests. This analysis looked the sub-question of “How is the US government using models, to what extent, how might this evolve, and what processes can prevent wide-scale adoption of widely available LLMs.” The original Google Doc can be found [here](#) and was posted online [here](#).

References

- AI as Normal Technology*, Retrieved July 8, 2025, from <http://knightcolumbia.org/content/ai-as-normal-technology>
- Artificial General Intelligence*, 2025. https://en.wikipedia.org/w/index.php?title=Artificial_general_intelligence&oldid=1301606989
- Cottier, B. *How Much Does It Cost to Train Frontier AI Models?*, 2024, June 3. <https://epoch.ai/blog/how-much-does-it-cost-to-train-frontier-ai-models>
- Crafts, N. Artificial Intelligence as a General-Purpose Technology: An Historical Perspective. *Oxford Review of Economic Policy*, 37(3), 521–536, 2021. <https://doi.org/10.1093/oxrep/grab012>
- Critch, A. *My Motivation and Theory of Change for Working in AI Healthtech*, 2024. <https://www.alignmentforum.org/posts/Kobbt3nQgv3yn29pr/my-motivation-and-theory-of-change-for-working-in-ai>
- Kulveit, J., Douglas, R., Ammann, N., Turan, D., Krueger, D., and Duvenaud, D. *Gradual Disempowerment: Systemic Existential Risks from Incremental AI Development*, 2025, January 29. <https://doi.org/10.48550/arXiv.2501.16946>
- Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Arx, S. V., Bloom, R., Broadley, T., Du, H., Goodrich, B., Jurkovic, N., Miles, L. H., Nix, S., Lin, T., Parikh, N., ... Chan, L. *Measuring AI Ability to Complete Long Tasks*, 2025, March 30. <https://doi.org/10.48550/arXiv.2503.14499>